

AltSchool Cybersecurity Capstone Project Report

Elastic SIEM SOC Deployment

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Executive Summary

For this project, I set up a small Security Operations Center (SOC) using Elastic SIEM. The goal was to collect logs from the cloud, use threat intelligence to detect suspicious activity, and practice investigating alerts. I also ran a Windows instance to do a simple forensic task with Event Viewer. Overall, the project helped me understand how a SOC detects and responds to incidents.

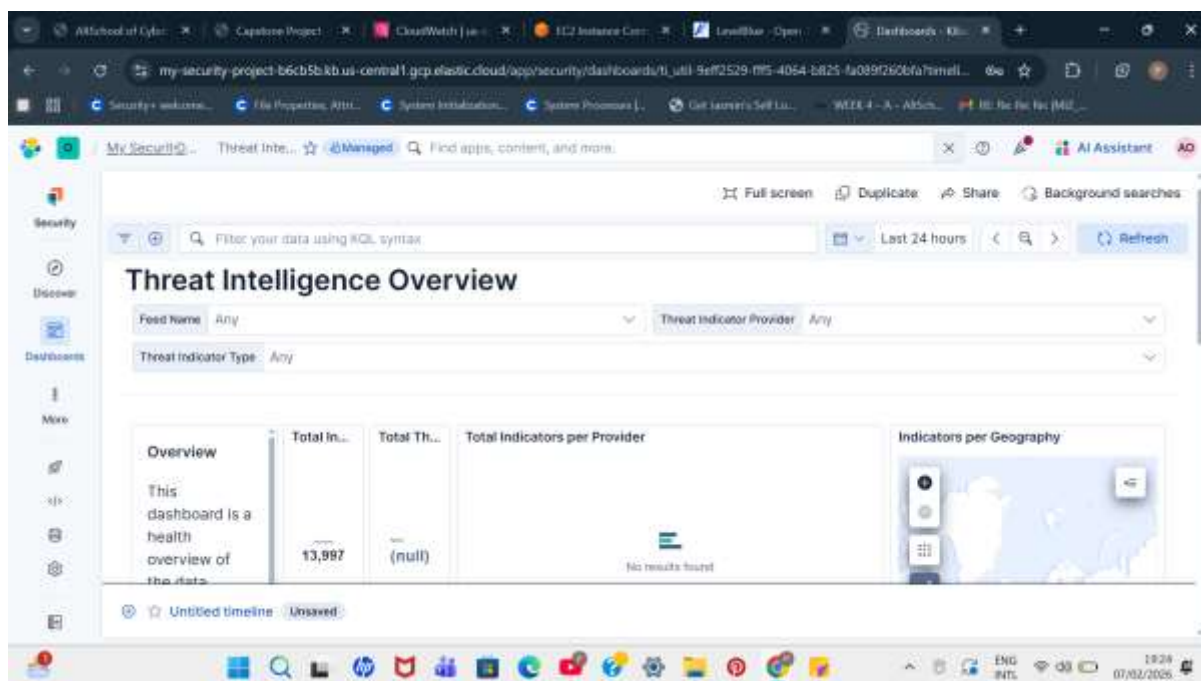
Phase 1: Environment Setup & Configuration

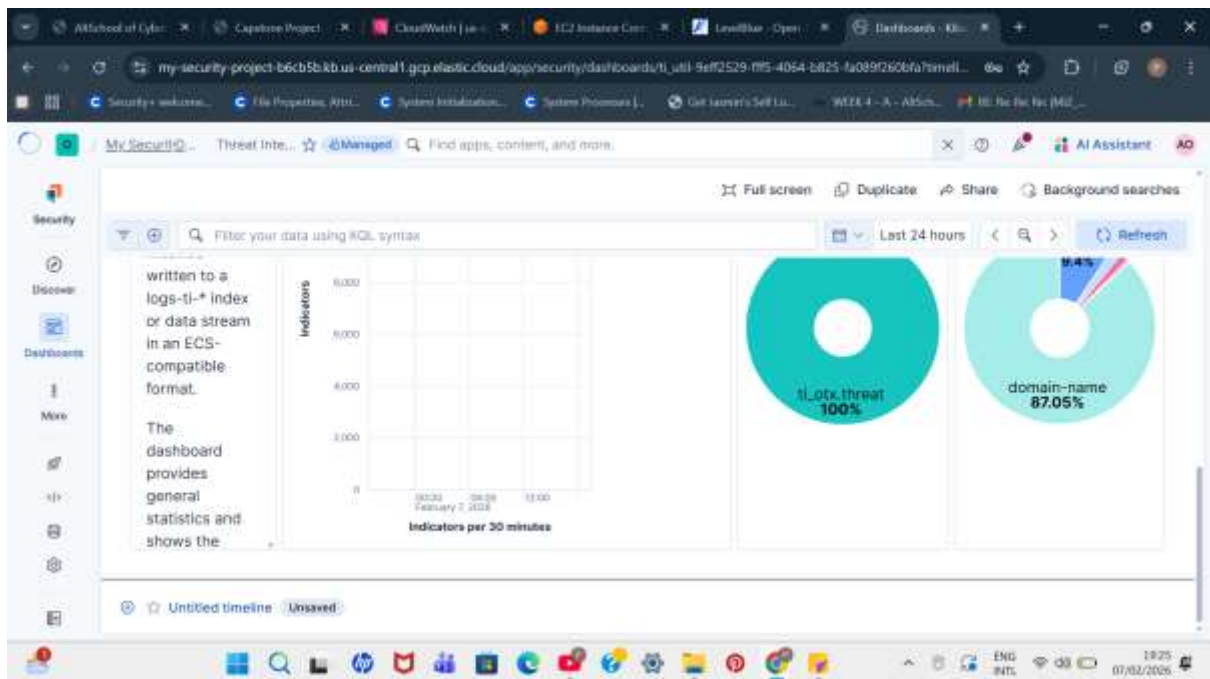
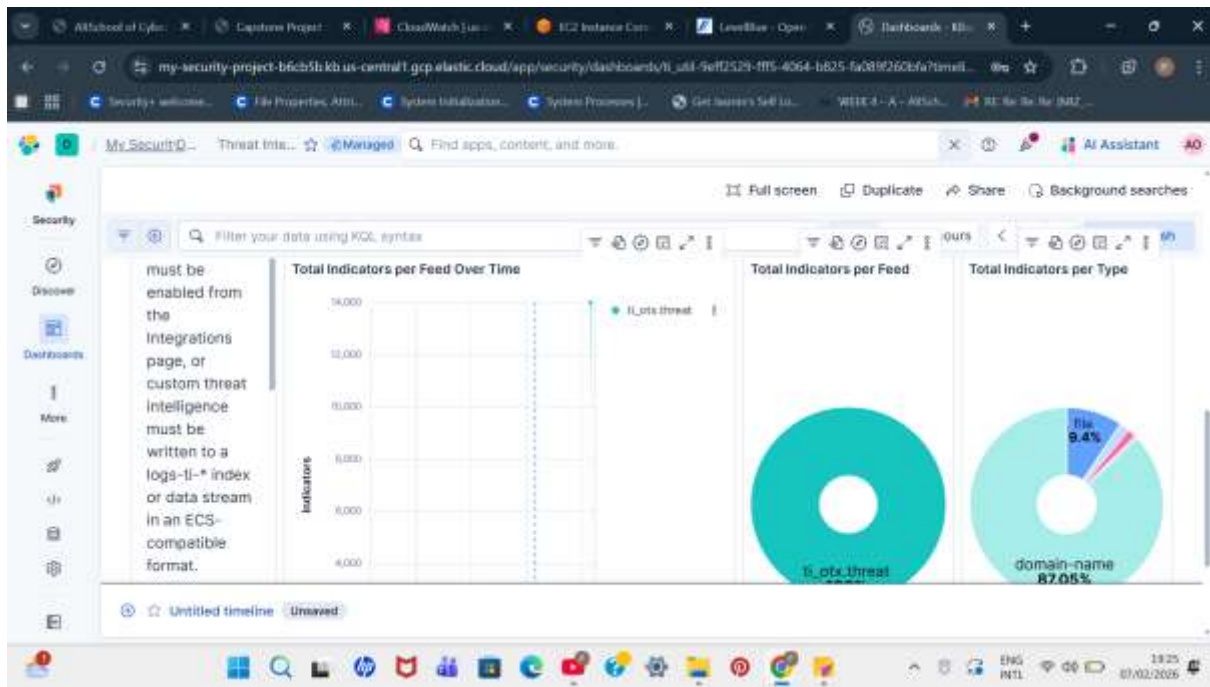
I deployed an Elastic SIEM instance and ingested logs from AWS, including EC2 instance logs, CloudTrail, and VPC Flow Logs.

I integrated AlienVault OTX for threat intelligence to enhance detection capabilities.

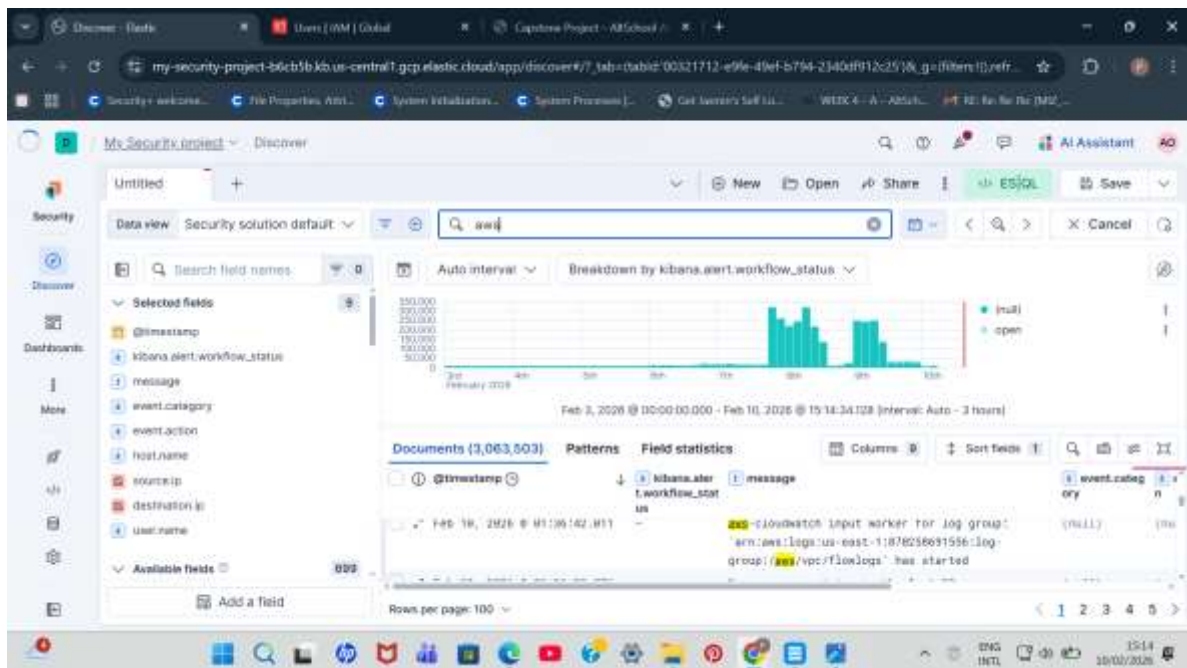
Evidence of functional ingestion and integration is shown in the screenshots below:

Appendix A: AlienVault OTX Integration overview.





Appendix B: Dashboard views showing active Flow Logs and Audit Logs.



Phase 2: Detection Engineering

When enabling the Out-of-the-Box (OOTB) detection rules, the following rules were available in my Elastic SIEM instance for AWS:

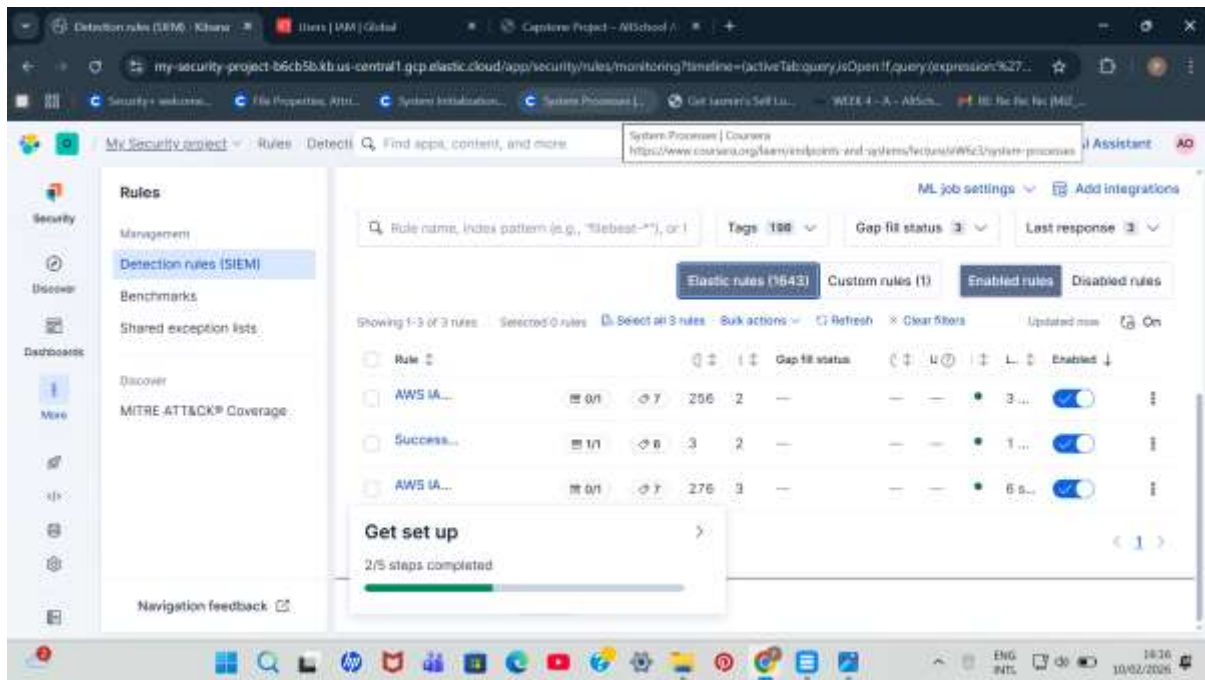
- AWS IAM Group Creation
- AWS IAM User Addition to Group
- SSH Activity / Successful SSH Authentication

Note: The rule “Unusual SSH Public Key Authentication” was not visible in my instance, so I enabled the available SSH-related detection rule, which monitors successful SSH logins.

Custom Rule:

- I created a custom detection rule to identify SSH traffic specifically on Port 22. This rule was successfully triggered during my simulation, validating detection capabilities.

Elastic SIEM Detections showing SSH authentication rule status:

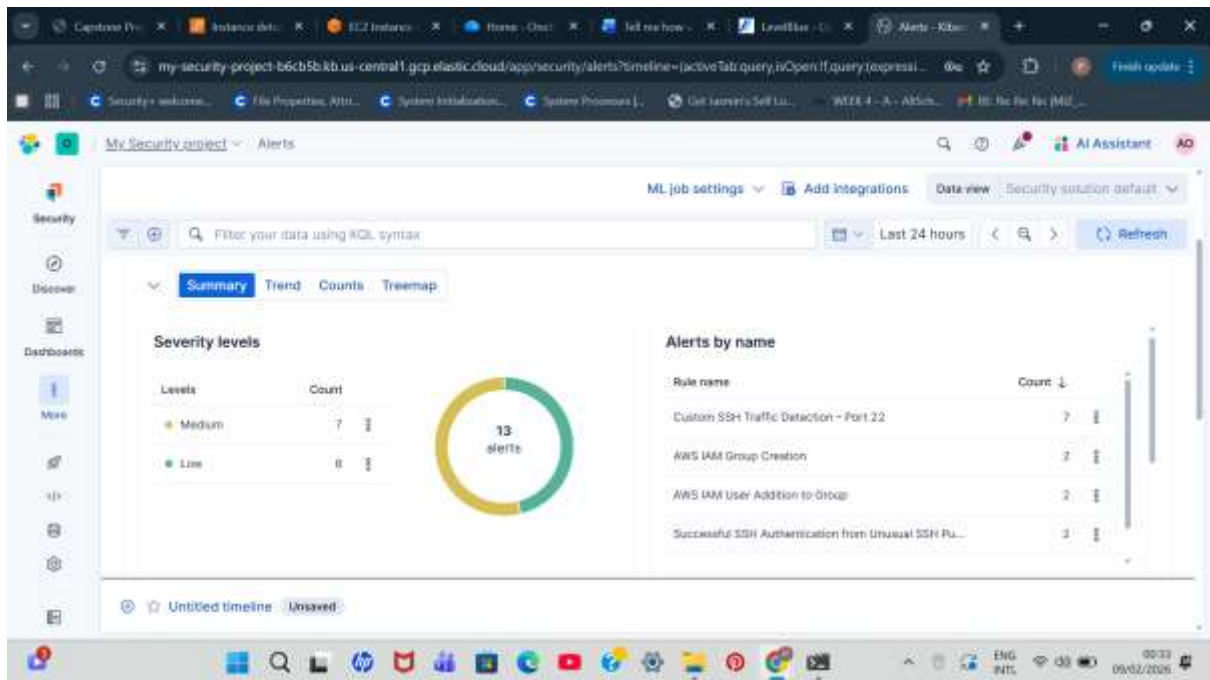


Phase 3: Attack Simulation & Validation

In this phase, I performed controlled actions on AWS to trigger the detection rules:

1. Creating a new IAM group.
2. Assigning administrative access to the group.
3. Creating a new IAM user.
4. Adding the new user to the group.
5. Establishing an SSH connection to the EC2 instance.

Elastic SIEM successfully detected these activities. The triggered alert captured in Appendix C shows the rule name, timestamp, and description: “Successful SSH Authentication from Unusual SSH Public Key.” This demonstrates that the SOC could effectively detect suspicious activity and respond to potential threats.



my-security-project-b6cb5bkb-us-central1-gcp-elasticcloud/app/security/alerts/timeline=activeTab:query,isActive:query,express...

My Security project Alerts

ML job settings Add integrations Data view Security solution default

Filter your data using RQL syntax Last 24 hours Refresh

Alert icon	Time	Rule name	Severity	Count	Description
	Feb 9, 2026 @ 00:27:18.957	Custom SSH Traffic Detecto...	medium	50	network event with sc
	Feb 9, 2026 @ 00:27:18.968	Custom SSH Traffic Detecto...	medium	50	network event with sc
	Feb 9, 2026 @ 00:27:18.965	Custom SSH Traffic Detecto...	medium	50	network event with sc
	Feb 9, 2026 @ 00:27:18.963	Custom SSH Traffic Detecto...	medium	50	network event with sc
	Feb 9, 2026 @ 00:24:55.144	Successful SSH Authentica...	low	21	authentication, sessio
	Feb 9, 2026 @ 00:22:25.139	AWS IAM User Addition to ...	low	21	iam event with source
	Feb 9, 2026 @ 00:22:25.137	AWS IAM User Addition to ...	low	21	iam event with source
	Feb 9, 2026 @ 00:20:43.033	AWS IAM Group Creation	low	21	iam event with source
	Feb 9, 2026 @ 00:20:43.030	AWS IAM Group Creation	low	21	iam event with source
	Feb 8, 2026 @ 04:27:18.276	Custom SSH Traffic Detecto...	medium	50	network event with sc
	Feb 8, 2026 @ 01:42:17.769	Custom SSH Traffic Detecto...	medium	50	network event with sc
	Feb 8, 2026 @ 01:37:17.838	Custom SSH Traffic Detecto...	medium	50	network event with sc

Untitled timeline Unsaved

Alerts - Ethers Detection rules (SIEM) - Kibana Users IAM Global Capstone Project - Michael C.

my-security-project-b6cb5b4b-us-central1-gcp-elastic-cloud/app/security/rules/id/0b0eda5b-b665-44de-af33-27c2bb1fab4f/alerts/timeline=activeTab...

Security > welcome... File Properties, Attributes System Initialization... System Processes... Get Jackson's Self-L... WELX 4 - A - Abs... IB: Re Re Re (Mid...

My Security project Rules Detection rules (SIEM) Successful SSH Authn... Alerts AI Assistant AO

Security

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More

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Rules

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Detection rules (SIEM)

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Shared exception lists

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MITRE ATT&CK® Coverage

ML job settings Add integrations Data view Security solution default

Filter your data using KQL syntax Last 30 days Refresh

Successful SSH Authentication fr...

Created by: 1788822089 on Feb 2, 2026 @ 21:38:12.480
Updated by: 1788822089 on Feb 7, 2026 @ 23:44:54.087

Last response: succeeded at Feb 10, 2026 @ 20:46:56.013 Notify when alerts generated

Enable Edit rule settings

About Details Investigation guide Setup guide

Get set up

2/5 steps completed

Definition

Index patterns logs-system.auth-*

Custom query alert.patterns.chained

ENG INTL 18:49 10/02/2026

Alerts - Elastic | Detection rules (SIEM) - Elastic | Users | IAM | Global | Capture Project - Michael F. | +

my-security-project-b6cb5b4b.us-central1-gcp.elastic.cloud/app/security/rules/id/0b0eda5b-b665-44da-af33-27c2bb1fab4f/alerts/timeline-(activeTab...

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Filter your data using KQL syntax

Last 30 days

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ABOUT

DESCRIPTION

This rule leverages the new Terms rule type to detect successful SSH authentications via a public key that has not been seen in the last 5 days. Public key authentication is a secure method for authenticating users to a server. Monitoring unusual public key authentication events can help detect

Author

Elastic

Get set up

2/5 steps completed

INDEX PATTERN

log-*system.auth*

CUSTOM QUERY

event.category:authentication and host.os.type:linux and event.action:ssh_login and event.outcome:success and system.auth.ssh.method:publickey

CUSTOM QUERY LANGUAGE

KQL

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my-security-project-b6cb5b4b.us-central1-gcp.elastic.cloud/app/security/rules/id/0b0eda5b-b665-44da-af33-27c2bb1fab4f/alerts/timeline-(activeTab...

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My Security project > Rules > Detection rules (SIEM) > Successful SSH Auth... > Alerts

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AUTHOR

Elastic

SEVERITY

Low

RISK SCORE

21

FALSE POSITIVE EXAMPLES

- This rule may trigger in cases where a user has routine work patterns that result in infrequent authentications.

LICENSE

Elastic License v2

Get set up

2/5 steps completed

CUSTOM QUERY LANGUAGE

KQL

RULE TYPE

New Terms

RELATED INTEGRATIONS

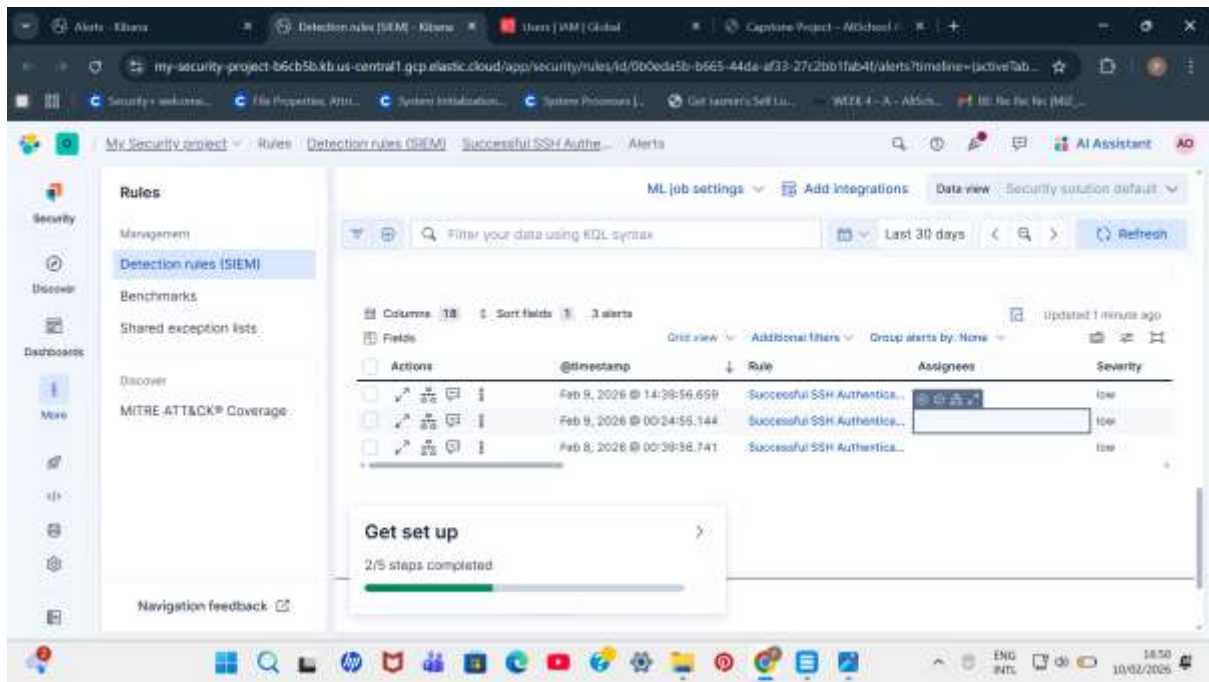
System > Installed

REQUIRED FIELDS

- event.action,
- event.category,
- event.outcome,
- host.os.type,
- system.auth.ssh.method

TIMELINE TEMPLATE

None

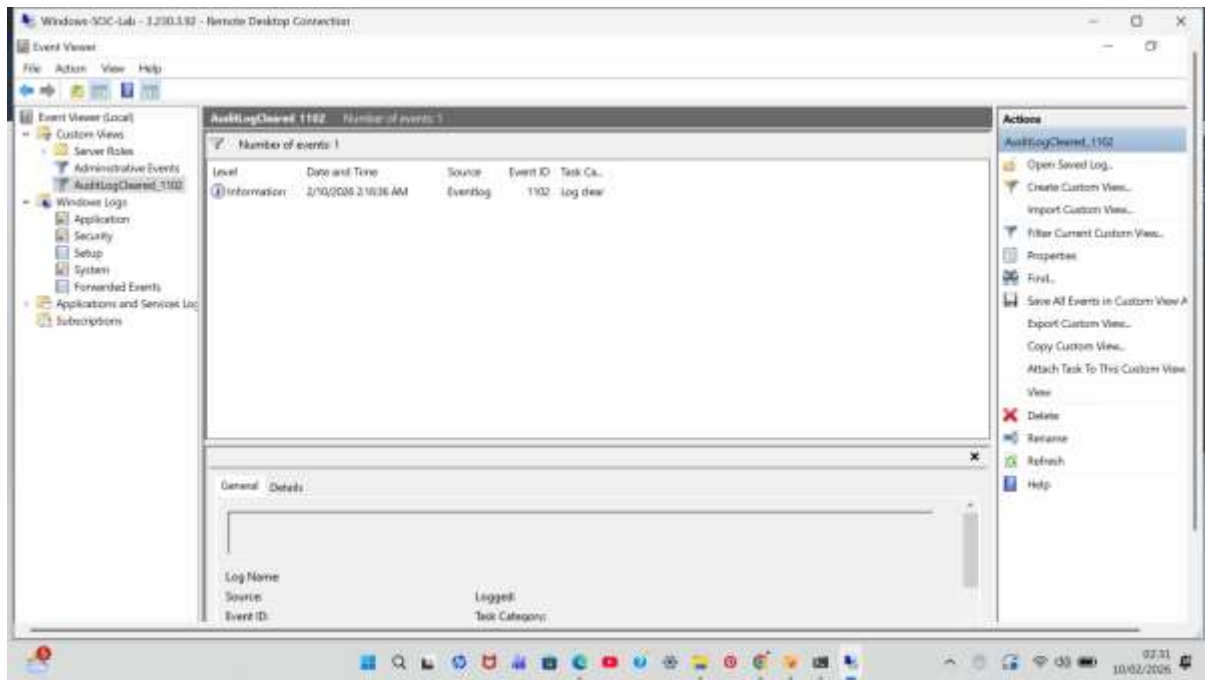


Phase 4: Endpoint Forensics (Windows)

For the endpoint forensics phase, I launched a Windows instance to monitor security log activity. Using Windows Event Viewer, I created a Custom View specifically for Event ID 1102, which indicates that the audit log was cleared.

I then cleared the security logs on the instance to trigger the event. Elastic's Event Viewer successfully recorded the activity, confirming that the Windows system captured the log-clearing event.

Appendix D shows Event ID 1102, including the timestamp and event details.



Incident Report

Timeline:

- IAM group and user created: Feb 2, 2026, 21:30–21:35
- SSH connection: Feb 2, 2026, 21:36
- Windows log cleared: Feb 10, 2026, 2:16:AM

Detection:

- Elastic SIEM detected the SSH connection via the custom rule.
- OOTB AWS rules detected IAM group/user modifications.
- Windows Event Viewer recorded Event ID 1102 when the audit log was cleared.

Triage:

- Threat level assessed as low, since the actions were controlled and intended for testing.
- Elastic SIEM successfully identified all test actions without false negatives.

Response:

- Recommended best practice is to continue monitoring for unusual SSH keys and audit log changes.
- For production, restrict administrative access and enforce multi-factor authentication.

Appendices

Appendix	Description
A	AlienVault OTX Integration overview screenshot.
B	Dashboard views showing active Flow Logs and Audit Logs.
C	Triggered OOTB and Custom SSH rule alerts (Phase 3).
D	Windows Event Viewer screenshot showing Event ID 1102 (Phase 4).

Critical Action Completed

- All cloud instances have been terminated.
- AWS integrations have been deleted.
- The SOC environment in Elastic retains the ingested logs for assessment.